

Symbol	Unit	Meaning
<i>Indices</i>		
i, o	Token	Input (sold) and output (bought) endpoint tokens, respectively; (i, o) is an ordered input–output pair and the direct route is $i \rightarrow o$.
k	Token	Candidate vehicle token used as the route intermediate; $k \notin \{i, o\}$.
ℓ, p	Token / pool	ℓ indexes tokens in cross-token sums; p indexes liquidity pools.
t, w	UTC day / UTC week	t indexes calendar days; w indexes calendar weeks in settlement variables.
τ	Days	Positive calendar-day horizon in dynamic specifications; the empirical panels use $\tau \in \{1, 7, 14, 30\}$.
q	USD	Input quote notional. Un-suffixed route-cost columns use $q = \$10,000$.
r	Route unit	Reconstructed input-to-output route unit. A coherent $i \rightarrow k \rightarrow o$ component contributes one r regardless of its number of legs; a split or join contributes one r per reconstructed input–output pair.
<i>Route and liquidity aggregates</i>		
$\text{Vol}_t, \text{DVol}_t$	USD	Vol_t is total realized USD volume across all direct and indirect route units on day t . DVol_t restricts it to direct routes, so $0 \leq \text{DVol}_t \leq \text{Vol}_t$.
$\text{IVol}_t, \text{IVol}_{k,t}$	USD	IVol_t restricts Vol_t to indirect routes, so $0 \leq \text{IVol}_t \leq \text{Vol}_t$. $\text{IVol}_{k,t}$ further restricts that volume to routes using vehicle k , so $0 \leq \text{IVol}_{k,t} \leq \text{IVol}_t$.
$N_t, N_{k,t}^{\text{in}}, N_{k,t}^{\text{out}}$	Route-unit count	N_t counts all reconstructed route units r on day t , not their individual legs. $N_{k,t}^{\text{in}}$ and $N_{k,t}^{\text{out}}$ restrict that count to routes with k as the input or output endpoint, respectively; each lies between 0 and N_t .
$N_t^I, N_{k,t}^I$	Route-unit count	N_t^I restricts N_t to route units with at least one intermediate token. $N_{k,t}^I$ further restricts that count to units with intermediate k , so $0 \leq N_{k,t}^I \leq N_t^I \leq N_t$; superscript I denotes indirect routes.
$\mathcal{A}_t, \mathcal{A}_t^k, \mathcal{M}_t^k$	Sets of token pairs	$\mathcal{A}_t$ is the set of endpoint pairs active in indirect routes on day t . $\mathcal{A}_t^k$ restricts it to pairs using k , and $\mathcal{M}_t^k$ further restricts it to pairs for which k has the largest candidate-vehicle volume; $\mathcal{M}_t^k \subseteq \mathcal{A}_t^k \subseteq \mathcal{A}_t$.
$\text{LegVol}_{k,t}^{\text{in}}, \text{LegVol}_{k,t}^{\text{out}}$	USD	$\text{LegVol}_{k,t}^{\text{in}}$ and $\text{LegVol}_{k,t}^{\text{out}}$ are route-leg USD volumes for which k is the leg’s input or output token, respectively, on day t .
$\text{Vol}_{k,t}^{\text{in}}, \text{Vol}_{k,t}^{\text{out}}$	USD	Input- and output-endpoint restrictions of Vol_t for token k : $0 \leq \text{Vol}_{k,t}^{\text{in}} \leq \text{Vol}_t$ and $0 \leq \text{Vol}_{k,t}^{\text{out}} \leq \text{Vol}_t$.
$\mathcal{K}$	Set of tokens	Prespecified candidate set $\{\text{WETH}, \text{USDC}, \text{USDT}, \text{DAI}, \text{WBTC}\}$.
$\mathcal{L}_{k,t}, m_p$	Set of pools / token count	$\mathcal{L}_{k,t}$ contains Uniswap V3 daily-snapshot pools with candidate k on one side, exact token contracts identified in the persisted V3 swap archive, and $0 < \text{TVL}_{p,t} \leq 10$ billion USD. m_p is the number of pool tokens in $\mathcal{K}$, so $m_p \in \{1, 2\}$.
$\text{TVL}_{p,t}, L_{k,t}$	USD	$\text{TVL}_{p,t}$ is pool p ’s day- t USD TVL from the V3 daily snapshot. $L_{k,t} = \sum_{p \in \mathcal{L}_{k,t}} \text{TVL}_{p,t}/m_p$; a one-candidate pool contributes all TVL to that candidate and a two-candidate pool contributes one half to each.
R_t^{WETH}	Daily log-return fraction	Day- t log return of the WETH price used to construct downside stress.
<i>Route-cost quote objects</i>		
$\mathcal{P}_{k,t,q}$	Set of token pairs	Quote-universe pairs for candidate k : distinct ordered input–output pairs among day t ’s 200 largest clean reconstructed pairs by realized USD volume, where clean means route class single or coherent , $k \notin \{i, o\}$, and each of i, o , and k has a valid day-price estimate. A valid estimate requires at least three finite token-side USD-per-token observations in (0, \$1,000,000) and equals their realized-USD-volume-weighted median. Each pair is submitted to the direct and via- k quote engines at input q ; membership does not require either quote to execute.
$\mathcal{D}_{k,t,q}, \mathcal{I}_{k,t,q}$	Sets of token pairs	$\mathcal{D}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ and $\mathcal{I}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ restrict the broad pair set to pairs with an executable direct route or an executable indirect route via k .
$\mathcal{C}_{k,t,q}$	Set of token pairs	Common-support pairs for candidate k , day t , and notional q : $\mathcal{C}_{k,t,q} = \mathcal{D}_{k,t,q} \cap \mathcal{I}_{k,t,q}$, so both the direct and indirect routes are executable.
$\mathcal{T}_{k,t,q}, \mathcal{W}_{k,t,q}$	Sets of token pairs	$\mathcal{T}_{k,t,q} \subseteq \mathcal{D}_{k,t,q}$ is the thin-direct subset. $\mathcal{W}_{k,t,q} \subseteq \mathcal{C}_{k,t,q}$ contains common-support pairs for which the indirect route beats the direct route, equivalently $\Delta C_{i,o,k,q,t}^D < 0$.
$D_{i,o,q,t}, I_{i,o,k,q,t}, T_{i,o,q,t}$	Indicator (0/1)	Indicators for direct-route availability (D), indirect-route-through- k availability (I), and an executable direct route returning less than $0.9q$ (T).
$O_{i,o,q,t}^D, O_{i,o,k,q,t}^I$	USD	Quoted output values; superscripts D and I denote direct and indirect routes.
$\Delta C_{i,o,k,q,t}^D$	Fraction	Pair-level direct cost advantage $(O_{i,o,q,t}^D - O_{i,o,k,q,t}^I)/O_{i,o,q,t}^D$ on $\mathcal{C}_{k,t,q}$; positive values favor the direct route.
<i>Settlement objects and operators</i>		
$\mathcal{R}_{k,w}$	Set of route units	Receipt-audited matched route units using vehicle k in UTC week w .
$\mathcal{R}_{k,w}^{\text{transfer}}$	Set of route units	Members whose receipt logs a transfer of vehicle k ; $\mathcal{R}_{k,w}^{\text{transfer}} \subseteq \mathcal{R}_{k,w}$.
$ \mathcal{S} , \mathbf{1}_{\{\cdot\}}$	Count / indicator (0/1)	Cardinality of any finite set $\mathcal{S}$ and an indicator equal to one when its subscripted condition is true.
<i>Dynamic operators</i>		
Δ_τ		Change in a daily variable over the τ days ending at t : $\Delta_\tau X_t = X_t - X_{t-\tau}$.

Variable	Formula	Unit	Data column	Definition
<i>Vehicle-use measures</i>				
$\text{VehicleShare}_{k,t}$	$\frac{\text{IVol}_{k,t}}{\text{IVol}_t}$	Fraction (0–1)	bridge_share	Fraction of day- t indirect-route USD volume routed through k .
$\text{AllRouteVehicleShare}_{k,t}$	$\frac{\text{IVol}_{k,t}}{\text{Vol}_t}$	Fraction (0–1)	all_route_bridge_share	Fraction of day- t all-route USD volume routed indirectly through k .
$\text{VehicleCountShare}_{k,t}$	$\frac{N_{k,t}^I}{N_t^I}$	Fraction (0–1)	bridge.count_share	Fraction of day- t indirect route units that use k .
$\text{PairCoverage}_{k,t}$	$\frac{ \mathcal{A}_t^k }{ \mathcal{A}_t }$	Fraction (0–1)	pair_coverage	Fraction of active indirect-route endpoint pairs that use k .
$\text{MainVehiclePairShare}_{k,t}$	$\frac{ \mathcal{M}_t^k }{ \mathcal{A}_t }$	Fraction (0–1)	pair_main_vehicle_share	Fraction of active indirect-route endpoint pairs for which k has the largest candidate-vehicle volume.
$\text{IVol}_{k,t}$		USD	bridge_volume.usd	Sum of realized USD volume over indirect routes using vehicle k on day t .
<i>Network and route-denominator controls</i>				
$\text{VolShare}_{k,t}$	$\frac{\text{LegVol}_{k,t}^{\text{in}} + \text{LegVol}_{k,t}^{\text{out}}}{\sum_{\ell} (\text{LegVol}_{\ell,t}^{\text{in}} + \text{LegVol}_{\ell,t}^{\text{out}})}$	Fraction (0–1)	vol_share	$\text{LegVol}_{k,t}^{\text{in}}$ and $\text{LegVol}_{k,t}^{\text{out}}$ are input-side and output-side route-leg USD volumes for k ; ℓ indexes every token in the day- t network.
$\text{Betweenness}_{k,t}$	$\frac{N_{k,t}^I}{N_t - N_{k,t}^{\text{in}} - N_{k,t}^{\text{out}}}$	Fraction (0–1)	betweenness.centrality	Fraction of day- t intent routes on which k is intermediate, excluding routes where k is the input or output endpoint. Superscripts identify each route role.
$\text{Betweenness}_{k,t}^{\text{vol}}$	$\frac{\text{IVol}_{k,t}}{\text{Vol}_t - \text{Vol}_{k,t}^{\text{in}} - \text{Vol}_{k,t}^{\text{out}}}$	Fraction (0–1)	volume.weighted.betweenness	USD-volume analogue of $\text{Betweenness}_{k,t}$ using the same route universe and input/output-endpoint exclusions.
Vol_t	$\text{DVol}_t + \text{IVol}_t$	USD	daily_all_route_volume.usd	Total realized USD volume across direct and indirect route units on day t .
IVol_t		USD	daily_indirect_route_volume.usd	Total realized USD volume across indirect route units on day t .
$\text{IndirectRouteShare}_t$	$\frac{\text{IVol}_t}{\text{Vol}_t}$	Fraction (0–1)	indirect_route_share	Fraction of day- t all-route USD volume executed through indirect routes.
<i>Liquidity measures</i>				
$L_{k,t}$	$\sum_{p \in \mathcal{L}_{k,t}} \frac{\text{TVL}_{p,t}}{m_p}$	USD	vehicle_linked_liquidity.usd	Candidate k ’s allocated share of valid Uniswap V3 pool TVL: full TVL when k is the pool’s only candidate token and one half when both pool tokens are candidates.
$\text{LPConc}_{k,t}$	$\frac{L_{k,t}}{\sum_{\ell \in \mathcal{K}} L_{\ell,t}}$	Fraction (0–1)	lp.concentration	Candidate k ’s share of total vehicle-linked liquidity on day t .
$\text{LogVehicleLiquidity}_{k,t}$	$\ln(1 + L_{k,t})$	Natural-log points	log_vehicle_linked_liquidity	Natural log of one plus $L_{k,t}$ measured in USD.
<i>Route-cost opportunity measures</i>				
$\text{DirectAvailable}_{k,t,q}$	$\frac{ \mathcal{D}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	direct_available_share	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $D_{i,o,q,t} = 1$. The un-suffixed data column fixes $q = \$10,000$.
$\text{IndirectAvailable}_{k,t,q}$	$\frac{ \mathcal{I}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	vehicle_available_share	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $I_{i,o,k,q,t} = 1$, so both legs through k are executable.
$\text{IndirectOnlyAvailable}_{k,t,q}$	$\frac{ \mathcal{I}_{k,t,q} \setminus \mathcal{D}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	no_direct_vehicle_available_share	Fraction of pairs with $D_{i,o,q,t} = 0$ and $I_{i,o,k,q,t} = 1$: no executable direct route but an executable route through k .
$\text{DirectDepth}_{k,t,q}$	$\text{median}_{\mathcal{D}_{k,t,q}} \left(O_{i,o,q,t}^D / q \right)$	Output USD per input USD	direct_depth.median	Median direct-route output ratio $O_{i,o,q,t}^D/q$ over executable-direct pairs $(i, o) \in \mathcal{D}_{k,t,q}$.
$\text{DirectCostAdvantage}_{k,t,q}$	$\text{median}_{\mathcal{C}_{k,t,q}} \Delta C_{i,o,k,q,t}^D$	Fraction	direct.cost_advantage.median	Median $\Delta C_{i,o,k,q,t}^D$ over common-support pairs $(i, o) \in \mathcal{C}_{k,t,q}$; positive values favor the direct route and negative values favor the indirect route through k .
$\text{IndirectBeatsDirect}_{k,t,q}$	$\frac{ \mathcal{W}_{k,t,q} }{ \mathcal{C}_{k,t,q} }$	Fraction (0–1)	vehicle_beats_direct.share	Fraction of $(i, o) \in \mathcal{C}_{k,t,q}$ for which $\Delta C_{i,o,k,q,t}^D < 0$, equivalently $O_{i,o,k,q,t}^I > O_{i,o,q,t}^D$.
$\text{ThinDirectShare}_{k,t,q}$	$\frac{ \mathcal{T}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	thin_direct_share	Fraction of pairs with an executable direct quote but $O_{i,o,q,t}^D/q < 0.9$. This is a quote-quality proxy, not a direct measure of pool liquidity depth.
<i>Stress and dynamic variables</i>				
Stress_t	$\max\{-R_t^{\text{WETH}}, 0\}$	Daily log-return fraction	stress_downside	Positive part of the negative day- t WETH log return.
StressEvent_t	$\mathbf{1}_{\{\text{Stress}_t \geq 0.08\}}$	Indicator (0/1)	stress_event_8pct	Equals one when Stress_t is at least 0.08 on day t .
$\Delta_\tau \text{VehicleShare}_{k,t}$	$\text{VehicleShare}_{k,t} - \text{VehicleShare}_{k,t-\tau}$	Fraction points	delta_bridge_share.t7	τ -day change for vehicle k ending on day t . The displayed data column is the $\tau = 7$ instance; the observations table also constructs $\tau \in \{1, 14, 30\}$.
<i>V4 settlement implementation measures</i>				
$\text{TransferIncidence}_{k,w}$	$\frac{ \mathcal{R}_{k,w}^{\text{transfer}} }{ \mathcal{R}_{k,w} }$	Fraction (0–1)	settlement_transfer_incidence	Fraction of receipt-audited route units $r \in \mathcal{R}_{k,w}$ containing an ERC-20 Transfer log for intermediate vehicle k .
$\text{ReceiptCount}_{k,w}$	$ \mathcal{R}_{k,w} $	Route-unit count	settlement_receipt_count	Number of receipt-audited matched route units for vehicle k in UTC week w .